

Systems of Linear Equations

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}$$

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$$a = 5$$

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Systems of Linear Equations

$$a + b + 3c = 15$$

$$a + 2b + 4c = 21$$

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$$\begin{array}{l} a + b + 3c = 15 \\ a + 2b + 4c = 21 \\ a + b + 2c = 13 \end{array} \quad \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \emptyset \\ \emptyset \\ \emptyset \end{bmatrix}$$

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Gaussian
Elimination

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \emptyset \\ \emptyset \\ \emptyset \end{bmatrix}$$

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$$r_2 = -r_1 + r_2$$

$$r_3 = -r_1 + r_3$$

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$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix} \quad \begin{array}{l} r_2 = -r_1 + r_2 \\ r_3 = -r_1 + r_3 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ -2 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ -2 \end{bmatrix}$$

Systems of Linear Equations

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 2 \end{bmatrix}$$

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$$a + b + 3c = 15$$

$$(5) + (4) + 3(2) = 15$$

Inverse Matrix A^{-1}



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$$A \cdot A^{-1} = I$$

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Inverse Matrix A^{-1}

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$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$(AB)^{-1} = A^{-1}B^{-1}$$

$$(A^T)^{-1} = (A^{-1})^T \quad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Inverse Matrix

$$A = \begin{bmatrix} 2 & 3 \\ 5 & 9 \end{bmatrix}$$

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$$A^{-1} = \frac{1}{18 - 15} \begin{bmatrix} 9 & -3 \\ -5 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ -5/3 & 2/3 \end{bmatrix}$$

Inverse Matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix}$$

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$$\therefore A A^{-1} = I$$

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